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Abstract

Magnetic Particle Imaging (MPI) is a high-sensitivity, non-invasive imaging modality with applications in stem cell tracking and pulmonary embolism detection, but its resolution is limited by the magnetic properties of existing tracer particles. Superferromagnetic iron oxide (SFMIO) nanoparticles demonstrate significantly sharper magnetic responses than traditional SPIOs, but require encapsulation in biocompatible droplets due to their solubility in nonpolar solvents. We developed a microfluidic droplet device to produce stable oil-in-water emulsions containing concentrated SFMIOS. Through a parametric study to minimize droplet size, minimize coalescence, and maximize droplet stability, a device thickness of 20 μm , no filters, treated with 5% polyvinyl alcohol (PVA), using a 2.5% concentration of sodium dodecyl sulfate (SDS), and a 1:2 oil-to-water flow ratio optimized the smallest droplet size with high emulsion stability. These emulsions are expected to improve MPI resolution and signal uniformity, enabling safer, radiation-free diagnostic imaging in future clinical applications.

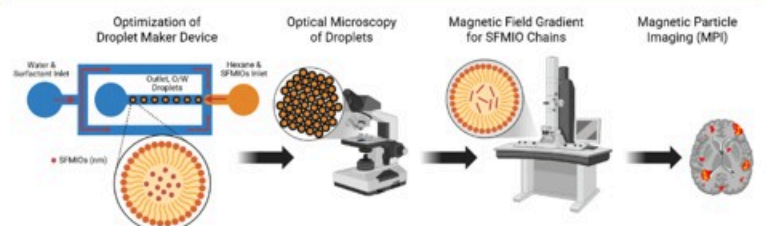


Figure 1. Schematic overview of experimental testing and parametric study.

Introduction

Magnetic Particle Imaging (MPI) is a non-invasive, radiation-free biomedical imaging modality that enables real-time, high-sensitivity visualization of magnetic nanoparticle tracers. Unlike conventional imaging methods such as MRI, PET, or CT, MPI directly detects the nonlinear magnetization of tracer particles under oscillating magnetic fields, resulting in high spatial and temporal resolution with no background tissue signal [1,2]. This makes MPI particularly useful for applications like stem cell tracking, pulmonary embolism detection, and cardiovascular imaging [3]. Traditional MPI tracers use superparamagnetic iron oxide nanoparticles (SPIOs), achieving $\sim 1\text{ mm}$ spatial resolution and $\sim 1\text{ }\mu\text{M}$ iron sensitivity [4].

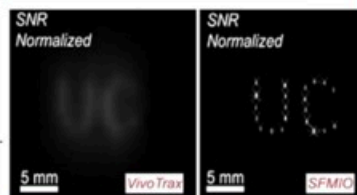
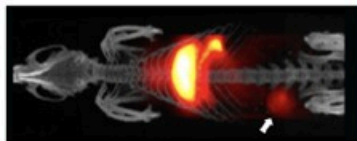


Figure 2. (2A) An MPI image of a tumor in a mouse model [12] & (2B) Phantom images comparing the resolution of standard MPI particles (Vivotrax) vs SFMIOS [13].

Recent advancements in tracer development have introduced superferromagnetic iron oxide nanoparticles (SFMIOs), which form magnetic chains under weak magnetic fields ($\sim 5\text{--}6\text{ mT}$) and demonstrate up to $28\times$ greater signal response and $38\times$ improved resolution compared to SPIOs [5,6]. Despite their magnetic advantages, SFMIOS are only soluble in nonpolar solvents such as hexane, making them incompatible with biological systems without additional encapsulation strategies [5,7]. To make these tracers clinically viable, SFMIOS must be embedded in biocompatible oil-in-water (O/W) emulsions that preserve their magnetic behavior and are safe for injection. However, encapsulating nanoparticles in microemulsions introduces new challenges, such as the formation of Pickering emulsions, where particles irreversibly adsorb to the oil-water interface, impairing droplet function and MPI responsiveness [8,9]. Prior work demonstrates that surfactant choice (e.g., SDS, Tween-80, Triton X-100) and concentration, device surface treatment (e.g., polyvinyl alcohol), and microfluidic flow parameters significantly affect emulsion stability and particle localization [10,11]. Microfluidic droplet generators provide a highly tunable platform to control droplet size, prevent coalescence, and ensure uniform particle loading, making them ideal for this application [12].

In this study, we designed a PDMS-based microfluidic droplet generator to produce $20\text{--}40\text{ }\mu\text{m}$ oil-in-water emulsions containing a high concentration of SFMIOS per droplet—sufficient to meet MPI's detection threshold of $\sim 1\text{ ng}/\mu\text{L}$ iron [6]. The device includes flow-focusing junctions, varying filter geometries, and hydrophilic surface treatment using 2% PVA to reduce nanoparticle adhesion. We conducted a parametric study by varying surfactant type (SDS, Tween-80, Pluronic F-68, Triton X-100), flow rate ratios (oil-to-water), and channel dimensions to minimize droplet size, maximize stability, and avoid Pickering emulsions. Preliminary results indicate successful droplet formation and encapsulation under multiple surfactant conditions, with droplets maintaining integrity for at least 5 minutes. This work lays the foundation for injectable, biocompatible MPI tracers, advancing the clinical translation of high-resolution, radiation-free imaging technologies.

Methods

To achieve a minimum droplet size with high emulsion stability, we model our parametric study using concentrated SPIOs in hexane on the following parameters that influence droplet diameter and formation in a droplet generator device:

Parameters:

- **Device Size:** Two devices with different heights $\times$ channel widths: $20\text{ }\mu\text{m} \times 20\text{ }\mu\text{m}$ and $40\text{ }\mu\text{m} \times 40\text{ }\mu\text{m}$.
- **Surface Treatment:** Half of our devices were treated with different concentrations of polyvinyl alcohol (PVA) (1%-5%) and the other half was left untreated.
- **Filter Type:** 4 filters were developed to avoid clumping of SPIOs in Hexane and ensure continuous flow throughout our device, highlighted in the Device & Fabrication section.
- **Surfactant and Concentration:** 4 surfactants were chosen based on high interfacial tension and previous research on droplet stability. Surfactants were diluted with distilled water with concentration values above the surfactant's critical micelle concentration (CMC). (1) Sodium dodecyl sulfate (SDS) (CMC = 8.2 mmol), (2) Tween-80 (CMC: $\sim 0.003\text{ mmol}$), (3) Pluronic F-68 CMC: $\sim 1\text{ mmol}$, and (4) Triton X-100 CMC: $\sim 0.3\text{ mmol}$ [10]
- **Flow Rate:** Experimentally determined ratios (hexane + SPIOs:water + surfactant) – 1:1, 1:2, 1:4, 1:8, etc.

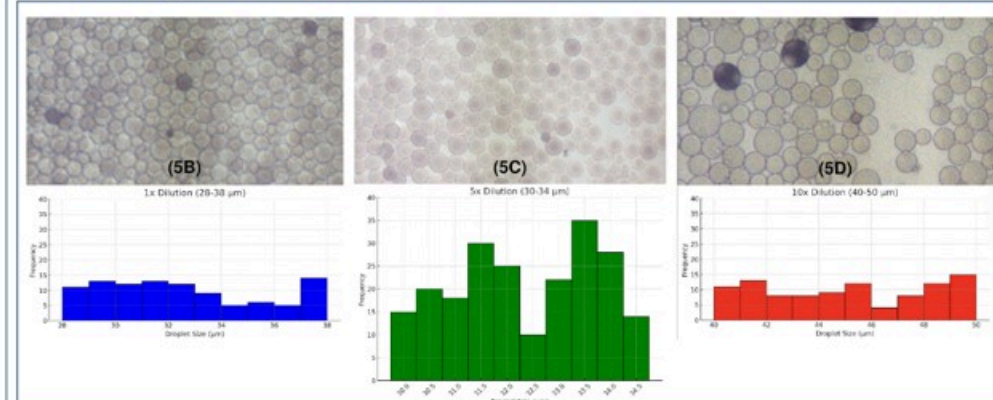
Measurements & Characterization:

- **Data Analysis:** When choosing parameters to test, we utilized a fractional Design of Experiments (DoE) to reduce the number of experiments by selecting key parameters to evaluate without testing all combinations of parameters.
- **Droplet Size Characterization:** O/W droplets were collected into a slide once droplets were stabilized at an optimal flow rate. Images were taken of the droplets and analyzed for droplet diameter and size distribution using ImageJ software.
- **Emulsion Stability:** Images of O/W droplets were taken periodically (every 10 minutes) to analyze for emulsion stability and the rate of droplet coalescence or evaporation.

Results

(5A) Droplet Size Characterization:
(5A) Droplets formed using SPIO-loaded hexane and 2.5% SDS in DI water at a 1:2 flow ratio (Flow rates: $40\text{ }\mu\text{L/hr}$ Hexane+SPIOs: $80\text{ }\mu\text{L/hr}$ DI water + SDS) produced the most stable and uniform droplets.
(5B-5D) Images showing how different levels of SPIO dilution affected droplet quality:
(5B) **1x dilution:** Very dense droplets with clumping due to high SPIO concentration and thickened oil phase. Droplets ($28\text{--}38\text{ }\mu\text{m}$) created with 1x dilution of SPIO concentration within hexane.
(5C) **5x dilution:** Best result - droplets were well-formed, consistent, and evenly distributed. Droplets ($30\text{--}34.5\text{ }\mu\text{m}$) created with 5x dilution of SPIO concentration within hexane.
(5D) **10x dilution:** Too little SPIO caused droplet collapse and uneven sizes, reducing stability. Droplets ($40\text{--}50\text{ }\mu\text{m}$) created with 10x dilution of SPIO concentration within hexane.

Figure 5. Oil-in-water emulsion droplets with concentrated SPIOs in hexane.



Emulsion Stability:

Using the 5x dilution of SPIOs in hexane, images were collected periodically to determine emulsion stability. Droplets were maintained up to 30 minutes before droplets started to coalesce and evaporate (Figure 6).

Figure 6. Periodic images of the 5x dilution droplets:

(6A) 0 mins, (6B) 10 mins, (6C) 20 mins, (6D) 30 mins.

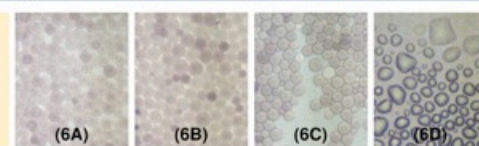
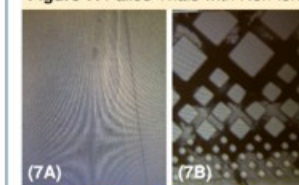


Figure 7. Failed Trials with Non-Ionic Surfactants and Different Filter Geometries



(7A) Attempts using Tween-20 and Triton X-100 as surfactants failed to generate discrete droplets. Continuous flow was observed in all trials, indicating incompatibility with SPIO-loaded hexane emulsions, likely due to insufficient interfacial tension control or surfactant-particle interaction.

(7B) Devices incorporating different filter geometries (e.g., wide grid, dense mesh) did not yield improvement in droplet uniformity or SPIO encapsulation. Filtered flow showed no observable enhancement over the baseline design, and in some cases, contributed to clogging or inconsistent flow profiles.

Best condition for minimal size and maximum emulsion stability: Device size: $20\text{ }\mu\text{m}$. Surface Treatment: 5% PVA, Filter type: No filter, Surfactant & Concentration: SDS 2.5%, Flow rate: 1:2 ($40\text{ }\mu\text{L/hr}$ SPIO-hexane: $80\text{ }\mu\text{L/hr}$ SDS-water).

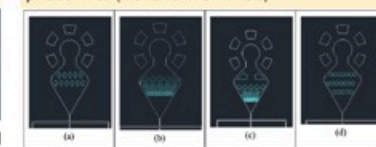
Device & Fabrication

Figure 3. Device design of microfluidic droplet generator device.



Continuous Phase (Water + Surfactant) Droplet Formation Junction Dispersed Phase (Hexane + SFMIOS)

Figure 4. Filter designs at the dispersed phase inlet (Hexane + SFMIOS)



Wide Grid Dense Mesh Chevron Circular Pillars

Table 1. Fabrication Process Overview

> Fabrication Parameters: (SU-8 2050 datasheet)

	Target Height: $20\text{ }\mu\text{m}$ Channel Width: $20\text{ }\mu\text{m}$	Target Height: $40\text{ }\mu\text{m}$ Channel Width: $40\text{ }\mu\text{m}$
Target Thickness (μm)	20	40
Photoresist Type	SU-8 2050	SU-8 2050
Spin Speed (RPM)	4000	3000
Pre-exposure Bake	65°C : 30 seconds 95°C : 4 minutes	65°C : 3 minutes 95°C : 4 minutes
Exposure Energy ($\frac{\text{mJ}}{\text{cm}^2}$)	165	160
UV Lamp Power ($\frac{\text{mW}}{\text{cm}^2}$)	10.75	10.75
Exposure Time (second)	13.48	14.88
Post-exposure Bake	65°C : 1 minute 95°C : 4 minutes	65°C : 1 minute 95°C : 4 minutes

Fabrication Key Steps:

- Wafer Cleaning
- SU-8 Spin Coating
- Photolithography
- Post-Exposure & Development
- Soft Lithography
- Inlets & Outlets Punching
- PDMS Bonding

Measured Dimensions:

- $20\text{ }\mu\text{m} \times 20\text{ }\mu\text{m}$ device:
 - $H = 32.93\text{ }\mu\text{m}$, $W = 22.35\text{ }\mu\text{m}$
- $40\text{ }\mu\text{m} \times 40\text{ }\mu\text{m}$ device:
 - $H = 45.70\text{ }\mu\text{m}$, $W = 44.69\text{ }\mu\text{m}$

Future Research & Conclusions

Optimizing Parameters & Emulsion Stability:

- The experimental research was performed utilizing traditional SPIOs rather than SFMIOS. The current study demonstrates the potential of using SDS as a surfactant to stabilize SFMIO-loaded emulsions, but further research is needed to refine the stability of SFMIO-loaded oil-in-water emulsions across varying surfactant concentrations. Future studies will focus on utilizing SFMIOS while identifying additional surfactants and their effects on droplet formation, size distribution, and MPI signal clarity.

Exploring Other Droplet Size Ranges:

- While the 5x dilution ($30\text{--}34\text{ }\mu\text{m}$) provides stable droplets for MPI applications, expanding the droplet size range through alternative formulation strategies may lead to improved resolution and a broader range of detectable sizes for medical imaging.

Flow Rate Optimization:

- The relationship between flow rate, surfactant concentration, and droplet formation is key to improving emulsion uniformity. Future work will focus on optimizing these parameters, potentially expanding the flow rate ranges tested and evaluating their effects on droplet stability and performance in MPI.

Advanced Device Design:

- Future work should explore modifications to the microfluidic device design, including alternative filter geometries and flow rates, to improve droplet uniformity and enhance the MPI signal, further optimizing the imaging performance.

Long-Term Stability:

- Investigating the long-term stability of SFMIO-loaded emulsions under physiological conditions will be crucial for evaluating their suitability in clinical settings, particularly for applications such as stem cell tracking and embolism detection.

References

- [1] Gleich B, Weizenecker J. Tomographic imaging using the nonlinear response of magnetic particles. *Nature*. 2005;435(7046):1214–7.
- [2] Weizenecker J, Gleich B, Rahmer J, et al. Three-dimensional real-time in vivo magnetic particle imaging. *Phys Med Biol*. 2009;54(5):L1–10.
- [3] Bulte JWM. Superparamagnetic iron oxides as MPI tracers: A primer and review of early applications. *Adv Drug Deliv Rev*. 2019;138:293–301.
- [4] Ferguson RM, Khandhar AP, Krishnan KM. Tracer design for magnetic particle imaging. *J Appl Phys*. 2012;111(7):07B318.
- [5] Tay ZW, Chandrasekharan P, Zhou XY, et al. Superferromagnetic nanoparticle chains for high-resolution magnetic particle imaging. *Small Methods*. 2021;5(8):2100180.
- [6] Fung KLB, Colson C, Bryan J, et al. First superferromagnetic remanence characterization and scan optimization for super-resolution magnetic particle imaging. *Nano Lett*. 2023;23(5):1717–25.
- [7] Zhou XY, Yu EY, Tay ZW, et al. Magnetic particle imaging for minimally invasive tracking of adipose-derived stem cells. *Theranostics*. 2017;7(16):4196–208.
- [8] Birks BP, Lumsdon SO. Pickering emulsions stabilised by monodisperse silica particles: Effect of particle size. *Langmuir*. 2001;17(10):2828–36.
- [9] Lebdioua K, Aimable A, Cerbelaud M, et al. Influence of different surfactants on Pickering emulsions stabilised by submicronic silica particles. *J Colloid Interface Sci*. 2018;520:127–33.
- [10] Lacava J, Quail AA, Raillard B, et al. On the behavior of nanoparticles in oil-in-water emulsions with different surfactants. *Soft Matter*. 2014;10(11):1696–704.
- [11] Teh SY, Lin R, Hung LH, Lee AP. Droplet microfluidics. *Lab Chip*. 2008;8(2):198–220.
- [12] Chandrasekharan P, Kuo R, Fung KLB, et al. Magnetic Particle Imaging in Vascular Imaging, Immunotherapy, Cell Tracking, and Noninvasive Diagnosis. *Molecular Imaging*. 2023;2023. doi:10.1155/2023/4131117
- [13] Colson C, Fung KLB, Bryan J, Tay ZW, Fellows BD, Saayuga C, Kuo R, Chandrasekharan P, Conolly SM. Evidence that SPIO chain formation is essential for high-resolution MPI. *bioRxiv*. 2022 Nov 28. <https://doi.org/10.1101/2022.11.27.518061>.